

Dhvani Dave

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Software Engineer who enjoys building meaningful products, solving problems and learning along the way.

EDUCATION

University at Buffalo

Master of Science in Computer Science; GPA: 3.9/4.0

Buffalo, NY

Aug. 2025 – Dec. 2026

Relevant Coursework: Algorithms, Machine Learning, Deep Learning, Operating Systems, Computer Security

University of Mumbai

Bachelor of Engineering in Computer Engineering; GPA: 3.6/4.0

Mumbai, India

Aug. 2019 – Jun. 2023

EXPERIENCE

Cloud Software Engineer

Jul. 2023 – Apr. 2025

CloudRock

Mumbai, India

- Built XML/XSLT data transformation pipelines in **Workday Studio** and custom EIB templates handling bulk loads of **10K+ records per run**, automating scheduled batch data exchange between Workday and downstream HR/Finance systems.
- Extended Workday integrations onto **AWS**, running the integration middleware on **EC2** and landing generated EIB and Studio output files in **S3**, with **Lambda** functions validating schema and routing before downstream HR/Finance systems consumed them.
- Implemented custom validation and business rule logic and wrote **Python** scripts to reconcile and pre-validate bulk data loads, reducing data errors by **30%** and cutting integration processing time.
- Designed and executed unit, integration, and regression test suites, and supported the post go-live hypercare phase by triaging and resolving production issues through **Jira** with cross-functional HR, Finance, and QA teams.

Software Engineer Intern

Aug. 2022 – May 2023

Vansh Consulting Inc.

Mumbai, India

- Developed an internal client-operations portal in **Python (Django)** and **React with TypeScript**, replacing spreadsheet-based tracking of supplier and purchase-order records across the consulting team.
- Built **REST APIs** and scheduled sync jobs that pulled order and vendor data from client ERP and CRM systems into **PostgreSQL**, adding indexes and rewriting slow joins to cut the nightly run from **~45 to ~15 minutes**.
- Containerized the application with **Docker** and set up a **GitHub Actions** pipeline for build, test, and deploy, writing ~60 unit and integration tests and replacing a manual release process that had caused repeated environment issues.
- Prototyped a support-ticket triage service in Python using **spaCy** and **scikit-learn** that classified incoming requests into 6 categories at **~85% accuracy**, reducing manual sorting for the helpdesk team.

PROJECTS

MiniDyn – Plugin-Based 3D Rigid Body Dynamics Simulator | C++, CMake, WebAssembly, Next.js

Jul. – Aug. 2026

- Built the engine in **C++** with a layered architecture: a **stable public API** over pluggable force elements and Baumgarte-stabilized joint constraints, so new physics models integrate without touching the solver core.
- Compiled the solver to WebAssembly to drive a **real-time Next.js dashboard** streaming energy and constraint-error telemetry at a **1 ms fixed timestep**, verified for numerical consistency by a GoogleTest suite in a CMake and GitHub Actions pipeline.

SentinelAI – Coral Hackathon | FastAPI, GraphQL, PostgreSQL, Neo4j

Apr. – May 2026

- Architected a **distributed 6-agent backend pipeline** (FastAPI) integrating PostgreSQL and Neo4j for **graph-based fraud detection**, exposing a GraphQL API consumed by the Next.js frontend for investigation reporting.
- Validated the pipeline end-to-end on a **155K-transaction** public dataset sample, surfacing high-risk transactions along with the **relationship paths** linking them, so an investigator could trace why each was flagged.

CodeSense – Agentic AI Developer Copilot | FastAPI, LangGraph, GCP, Qwen3-8B

Jan. – Mar. 2026

- Designed, coded, and deployed a **full-stack agentic debugging assistant** (FastAPI backend, React frontend) on **GCP** that retrieves code context, detects bugs, and generates validated patches across the full development lifecycle.
- Experimented with large language models (LLMs) by fine-tuning Qwen3-8B with QDoRA on 5,400 samples, improving **SWE-bench Token F1 from 0.075 to 0.257** and reducing hallucinations by **93%**.

TECHNICAL SKILLS

Languages: Python, C++, Go, Java, TypeScript, JavaScript, SQL

AI/ML: PyTorch, Hugging Face, scikit-learn, spaCy, NumPy, Pandas, LangChain, LangGraph, RAG, FAISS, Fine-Tuning (QDoRA)

Backend: FastAPI, Django, Node.js, REST, GraphQL | **Frontend:** React.js, Next.js, Streamlit

Data & Infra: PostgreSQL, MongoDB, Neo4j, AWS (EC2, S3, Lambda), GCP, Docker, Kubernetes, Linux

Tooling & Testing: Git, GitHub Actions, Tree-sitter, Jira

PROFESSIONAL CERTIFICATIONS

Building with the Claude API (*Anthropic*) | Introduction to Model Context Protocol (MCP) (*Anthropic*)

Advanced RAG with Vector Databases and Retrievers (*IBM*) | Machine Learning with Python (*IBM*)